

are part of the robot arm's built-in HMI, ensuring humans around it understand its state (active, paused, or error) at a glance. In essence, the robotic part itself is generating a micro-interaction separate from any central control room interface.

#### *Another everyday example.*

In high-end cars, a side mirror (a relatively peripheral 'part') can have a blind-spot warning LED on it, i.e., the mirror itself communicating to you.

**Smart parts increasingly demand user interaction or at least user awareness.**

**Multisensory interfaces.** These edge HMIs need not be full-screen; in fact, they shouldn't be, to avoid overwhelming complexity. Instead, multisensory micro-interfaces are used, such as lights, sounds, haptics, or even adaptive behaviour. A smart thermostat might have a simple colour glow that tells you whether it is heating or cooling without you having to read the numbers.

We are also seeing embedded HMI software being added to traditionally headless devices. Industrial controllers, for instance, used to always rely on a central SCADA or PC interface. Now, manufacturers are embedding miniature web servers or simple graphical displays in the device. A temperature controller or power meter might serve a small web dashboard or have a tiny screen so that an engineer can read data or configure it on the spot.

**The key point: User experience design is no longer just at the product level—it must be considered at the part and subsystem level.**

When developing an intelligent component, one should ask: How will the user (or maintainer or operator) know what this component is doing? For companies whose DNA is in part design and engineering, this is an opportunity to differentiate by offering HMI design as part of the part development process.

**Soon, along with the hardware and software datasheet for an intelligent module, such companies will increasingly deliver an HMI specification that tells the OEM's designers how this part should interact with users or other systems.**

### **The Tier-0.5 supplier: From component seller to end-user experience partner**

As products become a composition of many intelligent parts, no longer is a supplier just shipping a black-box component under specs; they are increasingly becoming co-creators of the end functionality. This has given rise to what the auto industry calls 'Tier 0.5' suppliers—those who sit between the traditional Tier-1 and the OEM, essentially acting as an extension of the OEM's development team.

In the automotive sector, the Tier-0.5 concept emerged as cars moved toward electric and autonomous technologies. OEMs found they needed much closer collaboration with tech suppliers to keep up with rapid innovation. In 1989, fewer than half of automotive suppliers contributed to part design; by 2011, about 70% were actively involved in part design for their OEM customers.

**Suppliers are not just building to print, they are jointly innovating. Tier-0.5 is a natural extension of that—a supplier deeply integrated in planning, design, and even identifying what the product's capabilities should be and experiences it can deliver.**

A Tier-0.5 supplier often provides a full subsystem or platform, rather than a single component. For example, consider the infotainment or 'digital cockpit' in a modern car. A Tier-1 vendor might traditionally supply a display and some electronics. A Tier-0.5 vendor, by contrast, might partner with the OEM to provide the entire cockpit computing platform—hardware, operating system, and

even an app ecosystem, customised to the automaker's brand needs. Case in point: a supplier named ECARX packaged an entire cockpit solution by bringing together multiple partners (chip designers, software firms like Epic Games for 3D engines, etc) and offered it as a ready-to-integrate package to OEMs. The OEM could then focus on tailoring the user experience and branding, rather than engineering the whole stack from scratch.

The Tier-0.5 dynamic is a bit different—it is about co-development of complex, high-tech elements of a product. Take the example we discussed of a smart RFID inlay or a sensor module with AI: an OEM could, in theory, develop the AI algorithms and analytics themselves, but if a supplier comes along with a ready-made intelligent module that not only captures data but also provides predictive alerts and an API for the OEM's app, that is tremendously valuable.

**The supplier in that case isn't selling just a component, they are selling expertise and a shortcut to functionality impacting end-user experience, effectively acting as a consultant and R&D partner for that aspect of the product. That's Tier-0.5 behaviour.**

For suppliers, stepping up to Tier-0.5 means expanding capabilities. It requires investing in R&D, software development, AI, and user experience—not just manufacturing. It also means earlier engagement with customers (at the product conception stage; not after specs are finalised) and new business models (ODM, shared IP or revenue, rather than simple per-unit pricing).

Electronics design and manufacturing firms recognise that evolving from Tier-1 to Tier-0.5 requires deeper technical conversations, perhaps even helping OEM clients set up global R&D centres (GCCs) or jointly dedicated innovation labs.

**The payoff is a stickier, more strategic relationship that is difficult**